

Multivariate (Joint) Distributions

ECON 97: Economic Toolkit

Patrick Molligo

UCLA

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<https://join.iclicker.com/RDNG>

Quiz 1

In an instant lottery game, the probability of drawing a winning ticket is 60%. If you buy 10 tickets, what's probability of having at least 5 winning tickets?

A. 0.9984

B. 0.9872

C. 0.8338

D. 0.0112

Table II *continued*

<i>n</i>	<i>x</i>	<i>p</i>									
		0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
8	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9984	0.9963	0.9922
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	0	0.6634	0.4305	0.2725	0.1678	0.1001	0.0576	0.0319	0.0168	0.0084	0.0039
	1	0.9428	0.8131	0.6572	0.5033	0.3671	0.2553	0.1691	0.1064	0.0632	0.0352
	2	0.9942	0.9619	0.8948	0.7969	0.6785	0.5518	0.4278	0.3154	0.2201	0.1445
	3	0.9996	0.9950	0.9786	0.9437	0.8862	0.8059	0.7064	0.5941	0.4770	0.3633
	4	1.0000	0.9996	0.9971	0.9896	0.9727	0.9420	0.8939	0.8263	0.7396	0.6367
	5	1.0000	1.0000	0.9998	0.9988	0.9958	0.9887	0.9747	0.9502	0.9115	0.8555
	6	1.0000	1.0000	1.0000	0.9999	0.9996	0.9987	0.9964	0.9915	0.9819	0.9648
9	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9983	0.9961
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	0	0.6302	0.3874	0.2316	0.1342	0.0751	0.0404	0.0207	0.0101	0.0046	0.0020
	1	0.9288	0.7748	0.5995	0.4362	0.3003	0.1960	0.1211	0.0705	0.0385	0.0195
	2	0.9916	0.9470	0.8591	0.7382	0.6007	0.4628	0.3373	0.2318	0.1495	0.0898
	3	0.9994	0.9917	0.9661	0.9144	0.8343	0.7297	0.6089	0.4826	0.3614	0.2539
	4	1.0000	0.9991	0.9944	0.9804	0.9511	0.9012	0.8283	0.7334	0.6214	0.5000
	5	1.0000	0.9999	0.9994	0.9969	0.9900	0.9747	0.9464	0.9006	0.8342	0.7461
	6	1.0000	1.0000	1.0000	0.9997	0.9987	0.9957	0.9888	0.9750	0.9502	0.9102
10	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9986	0.9962	0.9909	0.9805
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9980
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	0	0.5987	0.3487	0.1969	0.1074	0.0563	0.0282	0.0135	0.0060	0.0025	0.0010
	1	0.9139	0.7361	0.5443	0.3758	0.2440	0.1493	0.0860	0.0464	0.0233	0.0107
	2	0.9885	0.9298	0.8202	0.6778	0.5256	0.3828	0.2616	0.1673	0.0996	0.0547
	3	0.9990	0.9872	0.9500	0.8791	0.7759	0.6496	0.5138	0.3823	0.2660	0.1719
	4	0.9999	0.9984	0.9901	0.9672	0.9219	0.8497	0.7515	0.6331	0.5044	0.3770
	5	1.0000	0.9999	0.9986	0.9936	0.9803	0.9527	0.9051	0.8338	0.7384	0.6230
11	6	1.0000	1.0000	0.9999	0.9991	0.9965	0.9894	0.9740	0.9452	0.8980	0.8281
	7	1.0000	1.0000	1.0000	0.9999	0.9996	0.9984	0.9952	0.9877	0.9726	0.9453
	8	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9983	0.9955	0.9893
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9990
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	0	0.5688	0.3138	0.1673	0.0859	0.0422	0.0198	0.0088	0.0036	0.0014	0.0005
	1	0.8981	0.6974	0.4922	0.3221	0.1971	0.1130	0.0606	0.0302	0.0139	0.0059
	2	0.9848	0.9104	0.7788	0.6174	0.4552	0.3127	0.2001	0.1189	0.0652	0.0327
	3	0.9984	0.9815	0.9306	0.8389	0.7133	0.5696	0.4256	0.2963	0.1911	0.1133
	4	0.9999	0.9972	0.9841	0.9496	0.8854	0.7897	0.6683	0.5328	0.3971	0.2744
11	5	1.0000	0.9997	0.9973	0.9883	0.9657	0.9218	0.8513	0.7535	0.6331	0.5000
	6	1.0000	1.0000	0.9997	0.9980	0.9924	0.9784	0.9499	0.9006	0.8262	0.7256
	7	1.0000	1.0000	1.0000	0.9998	0.9988	0.9957	0.9878	0.9707	0.9390	0.8867
	8	1.0000	1.0000	1.0000	1.0000	0.9999	0.9994	0.9980	0.9941	0.9852	0.9673
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9993	0.9978	0.9941

Quiz 2

Let X denote the number of winning tickets. What's the mean of X ?

- A. 2
- B. 4
- C. 5
- D. 6

Review: Special Distributions

Binomial: 2 outcomes; probability p and $(1 - p)$; n trials

$$X \sim \text{bin}(n, p), \quad x = 0, 1, \dots, n$$

$$f(x) = {}_n C_x (p)^x (1 - p)^{(n-x)}$$

$$E[X] = np \quad \text{Var}(X) = np(1 - p)$$

Poisson: Count number of events over time/space; average rate λ

$$X \sim \text{Pois}(\lambda), \quad x = 0, 1, 2, \dots$$

$$f(x) = (e^{-\lambda}) \frac{\lambda^x}{x!}$$

$$E[X] = \text{Var}(X) = \lambda$$

Multiple random variables

Consider the following two random variables:

- $X = \text{Temperature}$
 - $X = 0$: Cold day
 - $X = 1$: Warm day
- $Y = \text{Ice cream sales}$
 - $Y = 0$: Low sales
 - $Y = 1$: Medium sales
 - $Y = 2$: High sales

Joint probability: prob. of X and Y simultaneously

E.g., $P(\text{Cold day} \cap \text{High sales}) = P(X = 0, Y = 2) = f(0, 2)$

Here, $f(x, y)$ is the “joint PMF”

Joint Probability Distribution

Tabular representation of the joint PMF:

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.30	0.15	0.05
$X = 1$	0.05	0.20	0.25

E.g., $P(X = 0, Y = 2) = 0.05$

Implies there is only 5% chance of high ice cream sales on cold days

All probabilities must still add up to 1

$$0.30 + 0.15 + 0.05 + 0.05 + 0.20 + 0.25 = 1$$

Marginal PMFs

Joint PMF: $f(x, y)$

Marginal PMFs: $f(x)$ and $f(y)$

The “marginal” probabilities represent individual random variables X, Y

Compute these by adding across the rows/columns of the PMF table

	$Y = 0$	$Y = 1$	$Y = 2$	
$X = 0$	0.30	0.15	0.05	0.5
$X = 1$	0.05	0.20	0.25	0.5
	0.35	0.35	0.3	1

$$P(X = 0) = P(X = 1) = 0.5$$

$$P(Y = 0) = 0.35 \quad P(Y = 1) = 0.35 \quad P(Y = 2) = 0.30$$

Conditional probability and independence

Calculate conditional probability in the usual way

$$P(Y = 2 \mid X = 1) = \frac{P(Y = 2, X = 1)}{P(X = 1)} = \frac{0.25}{0.05 + 0.20 + 0.25} = 0.50$$

Note that if X and Y are independent:

$$P(X, Y) = P(X)P(Y) \implies f(x, y) = f(x)f(y)$$

Must be true for all combinations of x, y

So we need marginal PMFs to prove independence

Mean/Variance of X and Y

Using the marginal PMFs, compute mean and variance: The mean of X is

$$E(X) = 0(0.50) + 1(0.50) = 0.50$$

$$E(Y) = 0(0.35) + 1(0.35) + 2(0.30) = 0.95$$

$$E(X^2) = 0^2(0.50) + 1^2(0.50) = 0.50$$

$$E(Y^2) = 0^2(0.35) + 1^2(0.35) + 2^2(0.30) = 1.55$$

$$\text{Var}(X) = 0.50 - (0.50)^2 = 0.25$$

$$\text{Var}(Y) = 1.55 - (0.95)^2 = 0.6475$$

Covariance of X and Y

Covariance: measures the relationship between 2 random variables.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y)$$

Covariance can be negative or positive.

- Negative: X, Y inversely related (E.g., temp. & snow shovel sales)
- Positive: X, Y positively related (E.g., temp. & ice cream sales)

If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

BUT, if $\text{Cov}(X, Y) = 0$, then X and Y might not be independent!

Covariance computation

$$E(XY) = \sum_x \sum_y (xy) f(x, y)$$

Using the joint PMF:

$$\begin{aligned} E(XY) &= (0)(0)(0.30) + (0)(1)(0.15) + (0)(2)(0.05) \\ &\quad + (1)(0)(0.05) + (1)(1)(0.20) + (1)(2)(0.25) = 0.70 \end{aligned}$$

We now have:

$$E(XY) = 0.70$$

$$E(X) = 0.50 \quad E(Y) = 0.95$$

Therefore,

$$\text{Cov}(X, Y) = 0.70 - (0.50)(0.95) = 0.225$$

Correlation of X and Y

Correlation is defined as

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

Covariance and correlation will have the same sign, but correlation is always between -1 and 1:

$$-1 \leq \rho_{X,Y} \leq 1$$

In our example:

$$\rho_{X,Y} = \frac{0.225}{\sqrt{0.25}\sqrt{0.6475}} \approx 0.559$$

Both $\rho_{X,Y} = -1$ and $\rho_{X,Y} = 1$ are called “perfect correlation”

Summary of joint random variables

Joint PMF: $f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$

Marginal PMF: $f(x_1) = P(X_1 = x_1)$ and $f(x_2) = P(X_2 = x_2)$

→ if X_1 & X_2 independent, then $f(x_1, x_2) = f(x_1)f(x_2)$

Covariance: $E(X_1 X_2) - E(X_1)E(X_2)$

where $E(X_1 X_2) = \sum_{x_1} \sum_{x_2} (x_1 x_2) f(x_1, x_2)$

Correlation: $\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} \in [-1, 1]$

→ if X_1 & X_2 independent, then covariance and correlation are 0

Linear functions of random variables

Consider random variables X_1, X_2

Means μ_1 and μ_2 ; variances σ_1^2 and σ_2^2 ; covariance σ_{12} ; correlation ρ_{12}

$$E[aX_1 \pm bX_2] = a\mu_1 \pm b\mu_2$$

$$\text{Var}(aX_1 \pm bX_2) = a^2\sigma_1^2 + b^2\sigma_2^2 \pm 2(ab)(\sigma_{12})$$

$$\sigma_{12} = \rho_{12}(\sigma_1\sigma_2)$$

If there are three variables:

$$\begin{aligned}\text{Var}(aX_1 \pm bX_2 \pm cX_3) &= a^2\sigma_1^2 + b^2\sigma_2^2 + c^2\sigma_3^2 \\ &\pm 2(ab)(\sigma_{12}) \pm 2(ac)(\sigma_{13}) \pm 2(bc)(\sigma_{23})\end{aligned}$$

Joint random variables practice

Let X_1 and X_2 have the joint PMF:

$$f(x_1, x_2) = c(x_1 + x_2), \quad x_1 = 1, 2 \quad x_2 = 1, 2$$

where c is a positive constant. Find the covariance between X_1 and X_2 .

Joint random variables practice